

State space models II

Version 1 (May 12, 2026)

ECON 31730, Spring 2026

University of Chicago

Department of Economics

Mikkel Plagborg-Møller

Estimating DSGE models

- A **Dynamic Stochastic General Equilibrium (DSGE)** model is a structural model of the macroeconomy driven by stochastic “shocks”.
- Given assumptions about agents’ preferences, technology, and information, the model yields equilibrium conditions that enforce optimal decisions and market clearing.
- We will see that many of these models can be (approximately) cast in linear state space model form. The state space coefficients are functions of deeper structural parameters.
- Assuming Gaussian shocks, get likelihood from Kalman filter.
- Given a prior, we can then do Bayesian inference. However, because the posterior does not have a simple closed form, we must rely on **Markov Chain Monte Carlo (MCMC)** methods.
- Great reference for all this: Herbst & Schorfheide (2015) book.

Example: simple New Keynesian model

- Simple log-linear model from Herbst & Schorfheide (2015), ch. 2.1.

$$\hat{y}_t = E_t(\hat{y}_{t+1}) - \frac{1}{\tau}[\hat{R}_t - E_t(\hat{\pi}_{t+1}) - E_t(\hat{z}_{t+1})] + \hat{g}_t - E_t(\hat{g}_{t+1})$$

$$\hat{\pi}_t = \beta E_t(\hat{\pi}_{t+1}) + \kappa(\hat{y}_t - \hat{g}_t)$$

$$\hat{R}_t = \rho_R \hat{R}_{t-1} + (1 - \rho_R)[\psi_1 \hat{\pi}_t + \psi_2(\hat{y}_t - \hat{g}_t)] + \varepsilon_{R,t}$$

$$\hat{g}_t = \rho_g \hat{g}_{t-1} + \varepsilon_{g,t}$$

$$\hat{z}_t = \rho_z \hat{z}_{t-1} + \varepsilon_{z,t}$$

- y_t : real GDP. π_t : gross inflation rate. R_t : nominal interest rate. g_t : government spending. z_t : transitory productivity growth rate.
- Hats denote log deviations from stochastic growth trend. Steady state inflation is some exogenous central bank target π^* .
- Exogenous i.i.d. Gaussian shocks: $\varepsilon_t \equiv (\varepsilon_{R,t}, \varepsilon_{g,t}, \varepsilon_{z,t})'$.

DSGE: log-linear equilibrium conditions

- Can write (log-linearized) DSGE equations in the canonical form

$$\Gamma_0 s_t = \Gamma_1 s_{t-1} + \Psi \varepsilon_t + \Pi \eta_t,$$

where s_t collects all model variables *as well as* expectations of endogenous variables.

- η_t are rational expectations forecast errors.
- E.g., in NK model from previous slide,

$$s_t \equiv (\hat{y}_t, \hat{\pi}_t, \hat{R}_t, \hat{g}_t, \hat{z}_t, E_t(\hat{y}_{t+1}), E_t(\hat{\pi}_{t+1}))',$$
$$\eta_t \equiv (\hat{y}_t - E_{t-1}(\hat{y}_t), \hat{\pi}_t - E_{t-1}(\hat{\pi}_t))'.$$

- Matrices Γ_0 , Γ_1 , Ψ , Π are functions of structural parameters θ .
- Note: $E_t(\hat{g}_{t+1})$ not included in s_t , because we can substitute $E_t(\hat{g}_{t+1}) = \rho_g \hat{g}_t$. Same for $E_t(\hat{z}_{t+1})$.

DSGE: stable solution

$$\Gamma_0 s_t = \Gamma_1 s_{t-1} + \Psi \varepsilon_t + \Pi \eta_t \quad (\dagger)$$

- Without any restrictions on forecast errors η_t , the system $(\dagger)$ may be explosive. Violates agents' optimality (transversality) conditions.
- The slide appendix provides conditions under which the system is non-explosive if

$$\eta_t = \tilde{\Psi} \varepsilon_t,$$

where $\tilde{\Psi}$ is *uniquely* given as a function of the coef matrices in $(\dagger)$.

- Substituting back into $(\dagger)$, we obtain a system of the form

$$s_t = \Theta_1(\theta) s_{t-1} + \Theta_\varepsilon(\theta) \varepsilon_t,$$

where the notation is a reminder that the coef matrices are ultimately functions of the structural parameters θ .

DSGE: state space representation and likelihood

$$s_t = \Theta_1(\theta)s_{t-1} + \Theta_\varepsilon(\theta)\varepsilon_t \quad (1)$$

- Above equations form the **transition equation** for state vector s_t .
- Some of the state variables are not directly observed in the data, such as the expectations $E_t(\hat{y}_{t+1})$, $E_t(\hat{\pi}_{t+1})$ and productivity $\hat{z}_t$.
- **Observation equation:** The data vector y_t^{data} satisfies

$$y_t^{\text{data}} = c + Hs_t + u_t, \quad (2)$$

where H will typically be a known matrix consisting of 1s and 0s. u_t may represent additional classical measurement error (indep. of ε_t).

- (1)–(2) constitute a standard linear state space model.
- Assuming ε_t and u_t are i.i.d. Gaussian, we can compute the likelihood for any structural parameters θ and noise variances.

Summarizing the posterior distribution

- Given a prior on the DSGE parameters, how do we compute the posterior distribution? Let's first take a step back.
- Consider a general statistical model with likelihood function $p(\mathcal{Y}_T | \theta)$ and prior density $\pi(\theta)$.
- Except in a few settings, the posterior density

$$p(\theta | \mathcal{Y}_T) = \frac{p(\mathcal{Y}_T | \theta)\pi(\theta)}{p(\mathcal{Y}_T)}, \quad p(\mathcal{Y}_T) = \int p(\mathcal{Y}_T | \theta) \pi(\theta) d\theta,$$

does not have a simple, known form (e.g., normal), so it's not straight-forward how to summarize the posterior distribution.

- Often marginal likelihood $p(\mathcal{Y}_T)$ is not available in closed form.

Generating draws from a target density

- If only we could generate *draws* $\theta^{(1)}, \dots, \theta^{(B)}$ from the posterior distribution, we could easily summarize it (e.g., mean, quantiles).
- In most applications, we are able to numerically compute the joint posterior *density* function (likelihood times prior density)

$$p(\theta \mid \mathcal{Y}_T) \propto p(\mathcal{Y}_T \mid \theta)\pi(\theta)$$

at any given θ , except for the proportionality constant $1/p(\mathcal{Y}_T)$.

- More general question (not necessarily about Bayesian inference):
 - ▶ Consider a **target density** $\psi(\theta) = C\tilde{\psi}(\theta)$, where $C > 0$.
 - ▶ We can compute $\tilde{\psi}(\theta)$ numerically for any θ . We may not know how to compute $C = (\int \tilde{\psi}(\theta) d\theta)^{-1}$.
 - ▶ How do we make draws from the distr. with density $\psi(\cdot)$?

Markov Chain Monte Carlo

- Many simulation-based techniques exist for generating draws from a target density. **Markov Chain Monte Carlo (MCMC)** methods are a generally applicable and diverse set of such techniques.
- Basic MCMC idea:
 - ① Construct an (entirely artificial) **Markov chain** on the parameter space Θ whose **stationary distribution** is the target density $\psi(\theta)$.
 - ② Do this by cleverly tinkering with the chain's **transition probabilities**.
 - ③ Given a starting draw $\theta^{(0)}$, simulate a long Markov chain $\theta^{(1)}, \dots, \theta^{(B)}$. After the chain has reached its stationary distribution, any further draws from the chain will be distributed according to the target density.
- Good econometrics reference: Chib (2001) handbook chapter.

Markov chains

- A (discrete time) **Markov chain** is a rule for (perhaps stochastically) stepping through elements of the set Θ . The next step is determined only by the current position.
- The **transition distribution** $P(\tilde{\theta} \mid \theta)$ is the distribution of the next draw $\tilde{\theta}$, given that the chain is currently at position θ .
- We will consider chains whose transitions are as follows:
 - ▶ Given current position θ , draw a **proposal** θ^* for the next step from continuous density $q(\theta^* \mid \theta)$.
 - ▶ With prob. $\alpha(\theta^* \mid \theta) \in [0, 1]$, let the next step equal the proposal: $\tilde{\theta} = \theta^*$.
 - ▶ With prob. $1 - \alpha(\theta^* \mid \theta)$, reject the proposal and stay at the current position: $\tilde{\theta} = \theta$.

Markov chains: stationary distribution

- The Markov chain has **stationary** (or invariant) density $\psi(\theta)$ if

$$\int P(\tilde{\theta} \in \mathcal{A} \mid \theta) \psi(\theta) d\theta = \int_{\mathcal{A}} \psi(\theta) d\theta$$

for any set $\mathcal{A} \subset \Theta$.

- In words, if θ is drawn from density $\psi(\theta)$, and we take one step with the Markov chain, then the new draw $\tilde{\theta}$ must also be distributed according to density $\psi(\theta)$.
- When the chain has reached the stationary distr., it stays there.
- It can be shown (see Chib), that the following **reversibility** condition is sufficient for $\psi(\theta)$ to be a stationary density:

$$q(\tilde{\theta} \mid \theta) \alpha(\tilde{\theta} \mid \theta) \psi(\theta) = q(\theta \mid \tilde{\theta}) \alpha(\theta \mid \tilde{\theta}) \psi(\tilde{\theta}) \quad \text{for all } \theta, \tilde{\theta}.$$

Intuitively, reversibility requires that flows from position θ to $\tilde{\theta}$ and vice versa are balanced under the stationary density.

Markov chains: stationary distribution (cont.)

- Under some additional *ergodicity* conditions (see Chib), a Markov chain – started at an arbitrary point $\theta^{(0)}$ – will converge to its stationary distribution (a.k.a. long-run distribution).
- Hence, after taking many initial steps along the Markov chain, each of the following draws will be very nearly distributed according to density $\psi(\theta)$.
- This is what we were after in the first place!
- So now we just need to figure out how to choose $q(\theta^* | \theta)$ and $\alpha(\theta^* | \theta)$ to ensure that our target density $\psi(\theta)$ is a stationary distribution for the chain.

Metropolis-Hastings algorithm

- Given the target density $\psi(\cdot)$ and some proposal density $q(\cdot \mid \cdot)$, the **Metropolis-Hastings** (MH) algorithm sets the **acceptance probability** equal to

$$\alpha(\theta^* \mid \theta) = \min \left\{ 1, \frac{\psi(\theta^*)q(\theta \mid \theta^*)}{\psi(\theta)q(\theta^* \mid \theta)} \right\}.$$

- Exercise:** Show that this $\alpha(\cdot \mid \cdot)$ guarantees that the reversibility condition holds, so $\psi(\cdot)$ is a stationary density of the chain.
- Get the desired stationary distr. *regardless* of proposal density $q(\theta^* \mid \theta)$. Can use this to fine-tune our algorithm (more soon).

Metropolis-Hastings algorithm (cont.)

- ① Pick some initial value $\theta^{(0)}$.
- ② For each iteration $n = 1, \dots, B$:
 - (i) Draw proposal $\theta^* \sim q(\theta^* \mid \theta^{(n-1)})$.
 - (ii) Draw $U \sim \text{unif}(0, 1)$.
 - (iii)
$$\theta^{(n)} = \begin{cases} \theta^* & \text{if } U \leq \frac{\psi(\theta^*)q(\theta^{(n-1)} \mid \theta^*)}{\psi(\theta^{(n-1)})q(\theta^* \mid \theta^{(n-1)})}, \\ \theta^{(n-1)} & \text{otherwise.} \end{cases}$$
- Algorithm only requires that we evaluate $\psi(\theta^*)/\psi(\theta)$ and $q(\theta^{(n-1)} \mid \theta^*)/q(\theta^* \mid \theta^{(n-1)})$.
- For Bayesian inference, don't need to compute proportionality constant $p(\mathcal{Y}_T) = \int p(\mathcal{Y}_T \mid \theta) \pi(\theta) d\theta$ (marginal likelihood)!

Random Walk Metropolis-Hastings

- Suppose we pick a symmetric proposal density:

$$q(\theta^* | \theta) = q(\theta | \theta^*) \quad \text{for all } \theta, \theta^*.$$

- Then the acceptance probability simplifies:

$$\alpha(\theta^* | \theta) = \min\{1, \psi(\theta^*)/\psi(\theta)\},$$

i.e., we always accept proposals that increase the target density relative to current position. But also sometimes accept proposals that decrease the density. Goal is not to *maximize* the density!

- Popular symmetric proposal:

$$(\theta^* | \theta) \sim N(\theta, \epsilon^2 I_k), \quad \text{i.e., } q(\theta^* | \theta) \propto e^{-\|\theta^* - \theta\|^2 / (2\epsilon^2)}.$$

Called **Random Walk Metropolis-Hastings**.

Example: Logit with normal prior

- Logit model of women's labor force participation (cross-sectional).
- Y_i : Indicator for woman's labor force participation.
- X_i : Constant and number of kids less than 6 years old (kidslt6).

$$P(Y = 1 | X) = \frac{\exp(X'\beta)}{1 + \exp(X'\beta)},$$

$$\beta \sim N(0, 5^2 I_2).$$

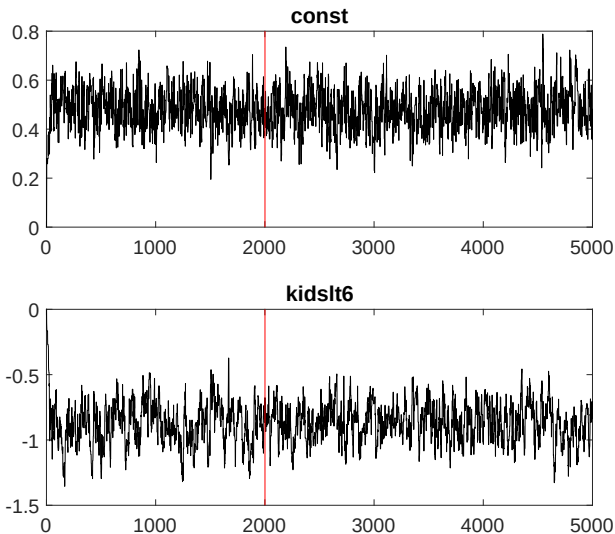
- Posterior is not available in closed form for this prior.

Example: Logit with normal prior

```
accepts = zeros(B,1);
draws = zeros(B,k);
log_targdens_curr = log_like(draws(1,:)) + log_prior(draws(1,:));
for n=2:B;
    prop = draws(n-1,:) + epsilon*randn(1,k);
    log_targdens_prop = log_like(prop) + log_prior(prop);
    if log(rand()) < log_targdens_prop - log_targdens_curr;
        accepts(n) = 1;
        draws(n,:) = prop;
        log_targdens_curr = log_targdens_prop;
    else
        draws(n,:) = draws(n-1,:);
    end;
end;
```

- Simulation parameters: $B = 5000$ total draws, step size $\epsilon = 0.1$.
- Discard first 2000 draws as **burn-in** when showing results.

Example: Logit with normal prior

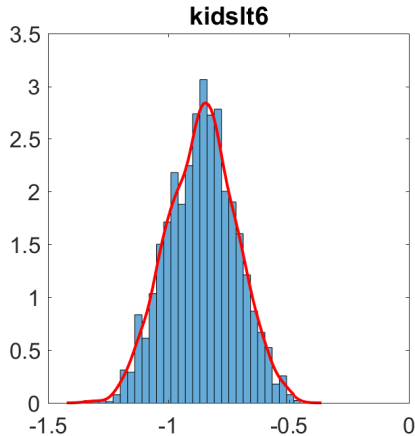
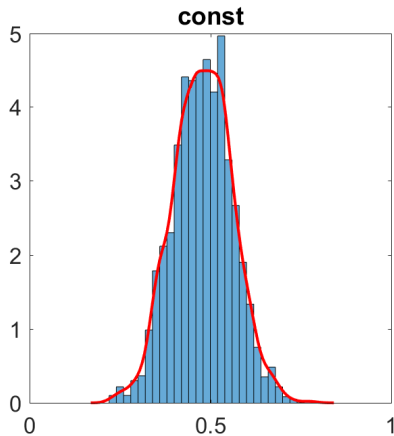


Trace plot of the Markov chain, 5000 total draws.
Red line marks end of chosen burn-in phase.

Example: Logit with normal prior

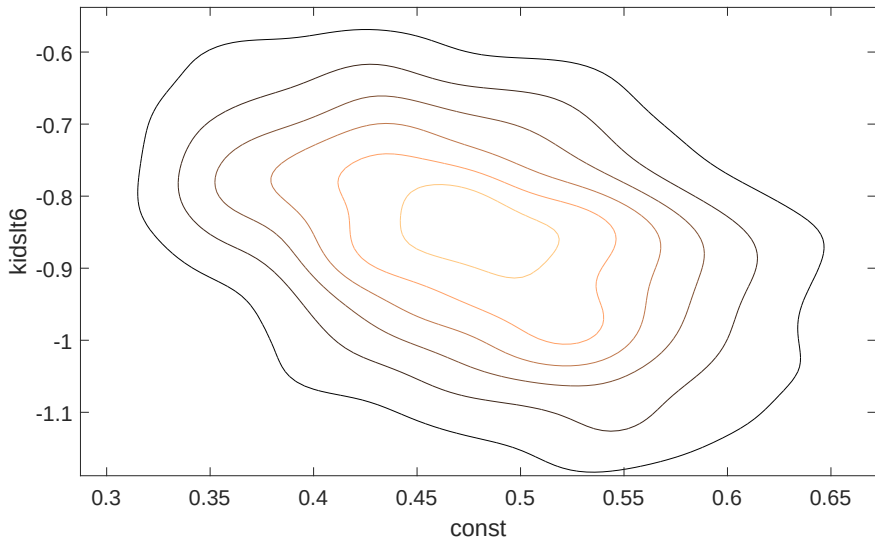
- How to determine when convergence has occurred?
- Should always eyeball the trace plot (previous slide).
- Should always start chains at various different initial values, run each chain for a long time, and compare the “long-run” distributions from each chain.
- More formal metrics based on comparing separate chains: Chib (2001), sec. 6.

Example: Logit with normal prior



Posterior draws: histogram and kernel density estimate.

Example: Logit with normal prior



Posterior draws: contour plot of joint density estimate.

Example: Logit with normal prior

- The Markov chain draws are not independent. In fact, they are almost always serially correlated, i.e.,

$$\rho_{j,\ell} \equiv \text{Corr}(\theta_j^{(n)}, \theta_j^{(n-\ell)}) \neq 0$$

for small $\ell \geq 1$. Typically $\rho_{j,\ell} > 0$ for small/moderate ℓ .

- As we know, under some regularity conditions,

$$\sqrt{B} \left(\frac{1}{B} \sum_{n=1}^B h(\theta_j^{(n)}) - E[h(\theta_j)] \right) \xrightarrow{d} N(0, \text{LRV}(h(\theta_j)))$$

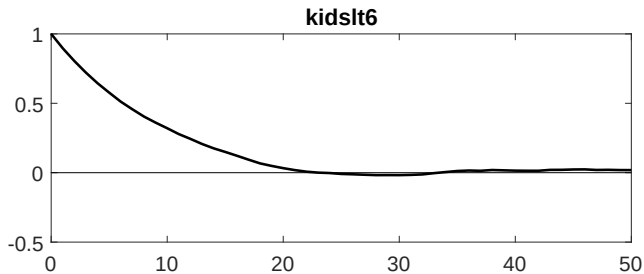
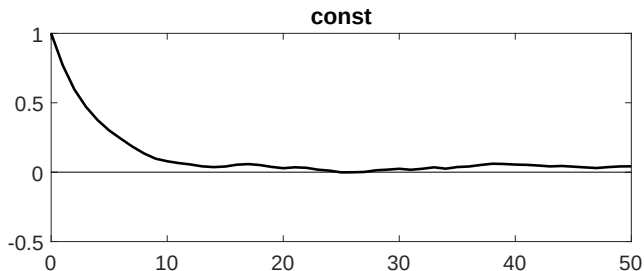
for any nice function $h(\cdot)$.

- The **effective sample size** is the Monte Carlo sample size that would have yielded the same estimator variance $\text{LRV}(h(\theta_j))/B$ in the ideal case where the draws are i.i.d.:

$$ESS_j = B \times \text{Var}(h(\theta_j)) / \text{LRV}(h(\theta_j)).$$

- Estimating ESS = estimating a LRV. See Chib sec. 3.2.

Example: Logit with normal prior



Plot of autocorrelations $\rho_{j,\ell}$. Horizontal axis: ℓ .

Example: Logit with normal prior

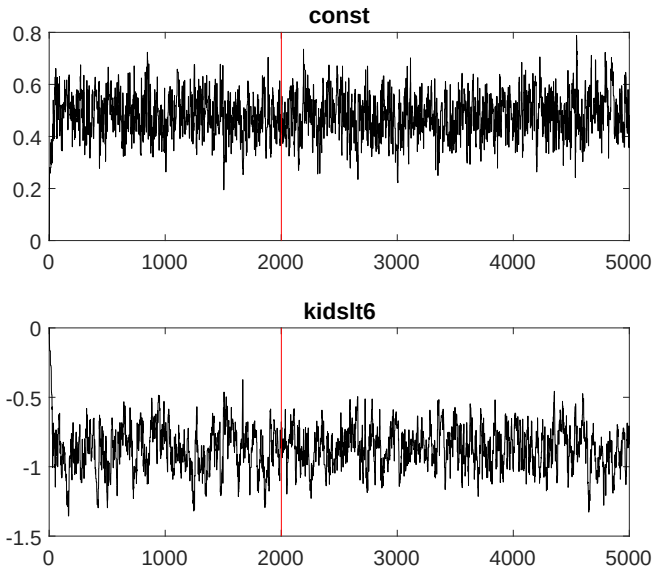
- The previous results used the proposal distribution

$$(\theta^* \mid \theta) \sim N(\theta, \epsilon^2 I_2)$$

with $\epsilon = 0.1$. The random walk **step size** ϵ determines the average magnitude of the Markov chain steps.

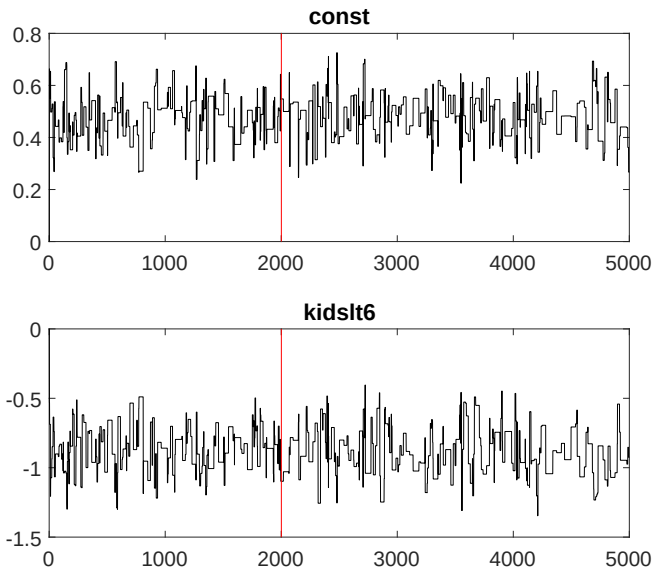
- With $\epsilon = 0.1$, the acceptance probability (after the first 2000 burn-in steps) is 0.55 in this example.
- Higher ϵ : Larger steps. We move more quickly around parameter space. But get smaller acceptance probability since most proposals will be in a direction of lower posterior density $\psi(\cdot)$.

Example: Logit with normal prior



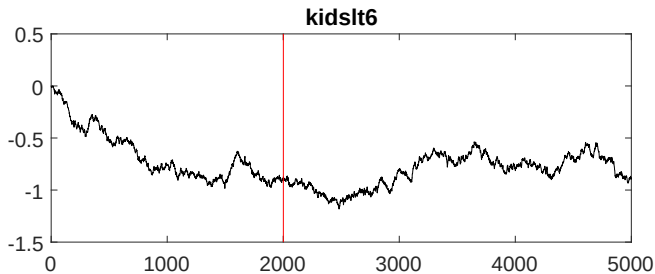
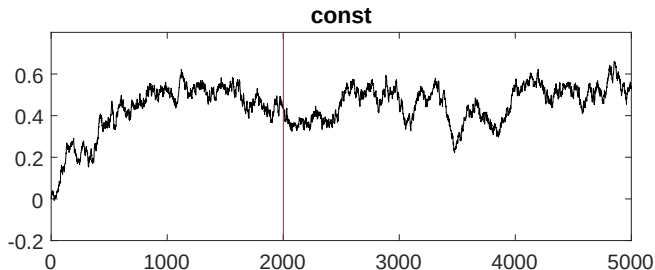
Trace plot, $\epsilon = 0.1$. Acceptance freq. = 0.55.

Example: Logit with normal prior



Trace plot, $\epsilon = 0.5$. Acceptance freq. = 0.07.

Example: Logit with normal prior



Trace plot, $\epsilon = 0.01$. Acceptance freq. = 0.95.

Proposal density

- We can choose any proposal density for the MH algorithm. Greatly influences the **mixing speed** of the Markov chain, i.e., how quickly we can generate relatively uncorrelated draws.
- In Random Walk proposals, often makes sense to tune step size ϵ to achieve acceptance frequency of about 0.3–0.6.
- Typically useful to have a different proposal std. dev. ϵ_j for each parameter θ_j . Could even have conditionally correlated proposals $(\theta^* | \theta) \sim N(\theta, \epsilon^2 \Sigma)$. Rule of thumb: $\Sigma \approx \text{Var}(\theta | \mathcal{Y}_T)$.
- Tuning the proposal density to the target is crucial in realistic problems. See Chib (2001) for common strategies.

Computing marginal likelihoods

- As we have discussed, the marginal likelihood

$$p(\mathcal{Y}_T) = \int p(\mathcal{Y}_T \mid \theta) \pi(\theta) d\theta$$

is useful for model selection.

- There exist general methods for numerically approximating the marginal likelihood using the draws from an MCMC procedure.
- See Herbst & Schorfheide (2015), ch. 4.6.

Estimating DSGE models in Dynare

- The open source software **Dynare** automates all stages of Bayesian DSGE estimation:
 - ▶ Log-linearizing the nonlinear equilibrium conditions.
 - ▶ Solving the model (state space representation).
 - ▶ Computing the likelihood.
 - ▶ Metropolis-Hastings.
 - ▶ Kalman smoothing of latent state variables.
- See the accessible “Dynare User Guide” (Mancini Griffoli, 2013).

Application: Smets & Wouters (2007)

TABLE 1A—PRIOR AND POSTERIOR DISTRIBUTION OF STRUCTURAL PARAMETERS

	Prior distribution			Posterior distribution			
	Distr.	Mean	St. Dev.	Mode	Mean	5 percent	95 percent
φ	Normal	4.00	1.50	5.48	5.74	3.97	7.42
σ_c	Normal	1.50	0.37	1.39	1.38	1.16	1.59
h	Beta	0.70	0.10	0.71	0.71	0.64	0.78
ξ_w	Beta	0.50	0.10	0.73	0.70	0.60	0.81
σ_l	Normal	2.00	0.75	1.92	1.83	0.91	2.78
ξ_p	Beta	0.50	0.10	0.65	0.66	0.56	0.74
ι_w	Beta	0.50	0.15	0.59	0.58	0.38	0.78
ι_p	Beta	0.50	0.15	0.22	0.24	0.10	0.38
ψ	Beta	0.50	0.15	0.54	0.54	0.36	0.72
Φ	Normal	1.25	0.12	1.61	1.60	1.48	1.73
r_π	Normal	1.50	0.25	2.03	2.04	1.74	2.33
ρ	Beta	0.75	0.10	0.81	0.81	0.77	0.85
r_y	Normal	0.12	0.05	0.08	0.08	0.05	0.12
$r_{\Delta y}$	Normal	0.12	0.05	0.22	0.22	0.18	0.27
$\bar{\pi}$	Gamma	0.62	0.10	0.81	0.78	0.61	0.96
$100(\beta^{-1} - 1)$	Gamma	0.25	0.10	0.16	0.16	0.07	0.26
$\bar{l}$	Normal	0.00	2.00	-0.1	0.53	-1.3	2.32
$\bar{\gamma}$	Normal	0.40	0.10	0.43	0.43	0.40	0.45
α	Normal	0.30	0.05	0.19	0.19	0.16	0.21

Note: The posterior distribution is obtained using the Metropolis-Hastings algorithm.

Source: Smets & Wouters (2007)

Application: Smets & Wouters (2007) (cont.)

TABLE 2—COMPARISON OF THE MARGINAL LIKELIHOOD OF
ALTERNATIVE VAR MODELS AND THE DSGE MODEL

Order of the VAR	No other prior	Sims and Zha (1998) prior
VAR(1)	−928.0	−940.9
VAR(2)	−966.6	−915.8
VAR(3)	−1018.1	−908.7
VAR(4)	−1131.2	−906.6
VAR(5)	—	−907.7
Memo: DSGE model	−905.8	−905.8

Note: In order to increase the comparability of the marginal likelihood of the various models, all models are estimated using the period 1956:1–1965:4 as a training sample (Sims 2003).

Source: Smets & Wouters (2007)

Application: Smets & Wouters (2007) (cont.)

TABLE 3—OUT-OF-SAMPLE PREDICTION PERFORMANCE

	GDP	dP	Fedfunds	Hours	Wage	CONS	INV	Overall
<i>VAR(1)</i>	<i>RMSE-statistic for different forecast horizons</i>							
1q	0.60	0.25	0.10	0.46	0.64	0.60	1.62	−12.87
2q	0.94	0.27	0.18	0.78	1.02	0.95	2.96	−8.19
4q	1.64	0.34	0.36	1.45	1.67	1.54	5.67	−3.25
8q	2.40	0.53	0.64	2.13	2.88	2.27	8.91	1.47
12q	2.78	0.63	0.79	2.41	4.09	2.74	10.97	2.36
<i>BVAR(4)</i>	<i>Percentage gains (+) or losses (−) relative to VAR(1) model</i>							
1q	2.05	14.14	−1.37	−3.43	2.69	12.12	2.54	3.25
2q	−2.12	15.15	−16.38	−7.32	−0.29	10.07	2.42	0.17
4q	−7.21	31.42	−12.61	−8.58	−3.82	1.42	0.43	0.51
8q	−15.82	33.36	−13.26	−13.94	−8.98	−8.19	−11.58	−4.10
12q	−15.55	37.59	−13.56	−4.66	−15.87	−3.10	−23.49	−9.84
<i>DSG</i>	<i>Percentage gains (+) or losses (−) relative to VAR(1) model</i>							
1q	5.68	2.05	−8.24	0.68	5.99	20.16	9.22	3.06
2q	14.93	10.62	−17.22	10.34	6.20	25.85	16.79	2.82
4q	20.17	46.21	1.59	19.52	9.21	26.18	21.42	6.82
8q	22.55	68.15	28.33	22.34	15.72	21.82	25.95	11.50
12q	32.17	74.15	40.32	27.05	21.88	23.28	41.61	13.51

Notes: All models are estimated starting in 1966:1. The forecast period is 1990:1–2004:4. VAR(1) and BVAR(4) models are reestimated each quarter, the DSGE model each year. The overall measure of forecast performance is the log determinant of the uncentered forecast error covariance matrix. Gains and losses in the overall measure are expressed as the difference in the overall measure divided by the number of variables and by two to convert the variance to standard errors (times 100).

Source: Smets & Wouters (2007)

Application: Smets & Wouters (2007) (cont.)

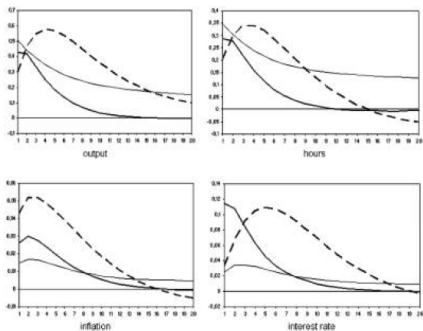


FIGURE 2. THE ESTIMATED MEAN IMPULSE RESPONSES TO "DEMAND" SHOCKS

Note: Bold solid line: risk premium shock; thin solid line: exogenous spending shock; dashed line: investment shock.

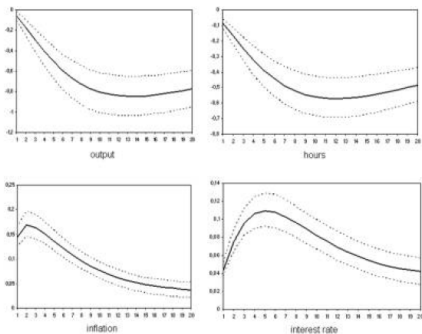


FIGURE 3. THE ESTIMATED IMPULSE RESPONSE TO A WAGE MARK-UP SHOCK

Note: The solid line is the mean impulse response; the dotted lines are the 10 percent and 90 percent posterior intervals.

Source: Smets & Wouters (2007)

List of other MCMC methods

- We have only scratched the surface of MCMC methods.
- In models with more than a few parameters, irregular likelihood functions, or nonlinearities, RWMH often works poorly.
- Usually advantageous to split the parameters into *blocks* with separate proposal densities. Alternate btw M-H steps for the separate blocks. Special case: **Gibbs sampling**, see Chib (2001).
- **Hamiltonian Monte Carlo** speeds up computations by using the gradient of the target density. See Neal (2011) handbook chapter.
 - ▶ Open-source software: Stan. Very user-friendly and well-documented.
- Herbst & Schorfheide (2015) provide an accessible introduction to the **particle filter** and **Sequential Monte Carlo**. Flexible and useful in nonlinear models and/or when the likelihood is multi-modal.

Appendix: Solving DSGE models

$$\Gamma_0 s_t = \Gamma_1 s_{t-1} + \Psi \varepsilon_t + \Pi \eta_t \quad (\dagger)$$

- Let $m \equiv \dim(s_t)$. **Generalized Schur decomposition:** There exist $m \times m$ matrices Q , Z , Λ , Ω such that

$$Q' \Lambda Z' = \Gamma_0, \quad Q' \Omega Z' = \Gamma_1, \quad Q Q' = Z Z' = I_m,$$

and Λ and Ω are upper-triangular.

- Define $w_t \equiv Z' s_t$. Pre-multiply $(\dagger)$ by Q :

$$\begin{pmatrix} \Lambda_{11} & \Lambda_{12} \\ 0 & \Lambda_{22} \end{pmatrix} \begin{pmatrix} w_{1,t} \\ w_{2,t} \end{pmatrix} = \begin{pmatrix} \Omega_{11} & \Omega_{12} \\ 0 & \Omega_{22} \end{pmatrix} \begin{pmatrix} w_{1,t-1} \\ w_{2,t-1} \end{pmatrix} + \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} (\Psi \varepsilon_t + \Pi \eta_t).$$

Appendix: Solving DSGE models (cont.)

$$\begin{pmatrix} \Lambda_{11} & \Lambda_{12} \\ 0 & \Lambda_{22} \end{pmatrix} \begin{pmatrix} w_{1,t} \\ w_{2,t} \end{pmatrix} = \begin{pmatrix} \Omega_{11} & \Omega_{12} \\ 0 & \Omega_{22} \end{pmatrix} \begin{pmatrix} w_{1,t-1} \\ w_{2,t-1} \end{pmatrix} + \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} (\psi_{\varepsilon_t} + \Pi \eta_t)$$

$$\implies w_{2,t} = \Lambda_{22}^{-1} \Omega_{22} w_{2,t-1} + \Lambda_{22}^{-1} Q_2 (\psi_{\varepsilon_t} + \Pi \eta_t)$$

- WLOG can re-order elements of s_t so eig.val's of $\Lambda_{22}^{-1} \Omega_{22}$ imply that $\{w_{2,t}\}$ is a purely explosive process. Define $r \equiv \dim(w_{2,t})$.
- Then the only way that the original system ($\dagger$) can be non-explosive is if $w_{2,0} = 0_{r \times 1}$ and

$$Q_2(\psi_{\varepsilon_t} + \Pi \eta_t) = 0_{r \times 1} \quad \text{for all } t.$$

- If $\dim(\eta_t) = r$ (the **Blanchard-Kahn condition**) and $Q_2 \Pi$ is invertible, then must have

$$\eta_t = -(Q_2 \Pi)^{-1} Q_2 \psi_{\varepsilon_t}.$$

- Details: Herbst & Schorfheide (2015), ch. 2.1.